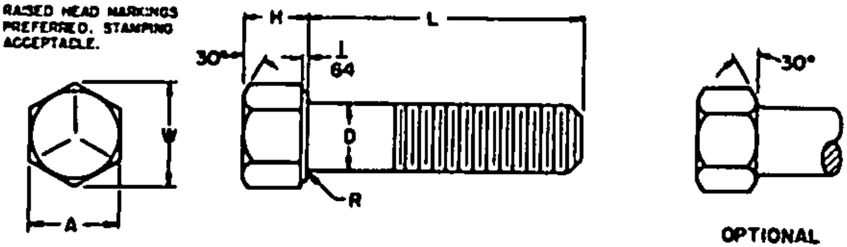
FED. SUP CLASS
5305

NOTE RAISED HEAD MARKINGS PREFERRED. STAMPING ACCEPTABLE.



			Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ
NOMINAL SIZE			1/4	5/16	3/8	7/16	1/2	9/16	5/8
THREADS PER INCH			20UNC	18UNC	16UNC	14UNC	13UNC	12UNC	11UNC
BODY DIAMETER	D	Max Min	.2500 .2450	.3125 .3065	.3750 .3690	.4375 .4305	.5000 .4930	.5625 .5545	.6250 .6170
WIDTH ACROSS FLATS	A	Max Min	.4375 .4280	.5000 .4890	.5625 .5510	.6250 .6120	.7500 .7360	.8125 .7950	.9375 .9220
WIDTH ACROSS CORNERS	W	Max Min	.5050 .4980	.5770 .5570	.6500 .6250	.7220 .6980	.8660 .8400	.9380 .9100	1.0830 1.0510
HEAD HEIGHT	H	Max Min	.1630 .1500	.2110 .1950	.2430 .2260	.2910 .2720	.3230 .3020	.3710 .3480	.4030 .3750
RADIUS	R	Max Min	.023 .009	.023 .009	.023 .009	.023 .009	.023 .009	.041 .021	.041 .021

LENGTH L	Tolerance	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/8	+0 -1/32	~~1~~	~~27~~	~~53~~				
7/16		~~2~~	~~28~~	~~54~~	~~79~~			
1/2		~~3~~	29	55	~~80~~	~~104~~		
9/16		~~4~~	~~30~~	~~56~~	~~81~~	~~105~~	~~130~~	
5/8		~~5~~	~~31~~	57	~~82~~	~~106~~	~~131~~	~~155~~
3/4		~~6~~	~~32~~	~~58~~	~~83~~	~~107~~	~~132~~	~~156~~
7/8		~~7~~	~~33~~	~~59~~	~~84~~	~~108~~	~~133~~	~~157~~
1		~~8~~	~~34~~	~~60~~	~~85~~	~~109~~	~~134~~	~~158~~
1-1/8	+0 -1/16	~~9~~	~~35~~	~~61~~	~~86~~	~~110~~	~~135~~	~~159~~
1-1/4		~~10~~	~~36~~	62	~~87~~	~~111~~	~~136~~	~~160~~
1-3/8		~~11~~	~~37~~	~~63~~	~~88~~	~~112~~	~~137~~	~~161~~
1-1/2		~~12~~	~~38~~	64	~~89~~	~~113~~	~~138~~	~~162~~
1-3/4		~~13~~	~~39~~	~~65~~	~~90~~	~~114~~	~~139~~	~~163~~
2		~~14~~	~~40~~	66	~~91~~	~~115~~	~~140~~	~~164~~
2-1/4	+0 -1/32	~~15~~	~~41~~	67	~~92~~	~~116~~	~~141~~	~~165~~
2-1/2		~~16~~	~~42~~	68	~~93~~	~~117~~	~~142~~	~~166~~
2-3/4		~~17~~	~~43~~	~~69~~	~~94~~	~~118~~	~~143~~	~~167~~
3		~~18~~	~~44~~	70	~~95~~	~~119~~	~~144~~	~~168~~
3-1/4		~~19~~	~~45~~	71	~~96~~	~~120~~	~~145~~	~~169~~
3-1/2		~~20~~	~~46~~	72	~~97~~	~~121~~	~~146~~	~~170~~
3-3/4		~~21~~	~~47~~	73	~~98~~	~~122~~	~~147~~	~~171~~
4		~~22~~	~~48~~	74	~~99~~	~~123~~	~~148~~	~~172~~
4-1/4		~~23~~	~~49~~	~~75~~	~~100~~	~~124~~	~~149~~	~~173~~
4-1/2		~~24~~	~~50~~	76	~~101~~	~~125~~	~~150~~	~~174~~
4-3/4		~~25~~	~~51~~	~~77~~	~~102~~	~~126~~	~~151~~	~~175~~
5		~~26~~	~~52~~	78	~~103~~	~~127~~	~~152~~	~~176~~
5-1/2						~~128~~	~~153~~	~~177~~
6						~~129~~	~~154~~	~~178~~

MATERIAL: Medium Carbon Steel, Grade 5 of procurement specification. Minimum tensile strengths: 120,000 psi for diameters through 3/4"; 110,000 psi for 7/8" and 1" diameters; 105,000 psi for diameters 1-1/8" through 1-1/2"; 90,000 psi for diameters over 1-1/2".

FINISH: Plain (untreated).
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.
MINIMUM THREAD LENGTH: Shall be twice the diameter plus 1/4" for lengths up to and including 6". The tolerance shall be plus 3/16" or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, for sizes up to and including 1" and 3-1/2 threads for sizes larger than 1".
NOTES:
(1) Referenced documents of issue in effect on date of invitation for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example: MS35295-1.
(4) All dimensions in inches.

Ⓑ CANCELLED IN PART
SEE SHEET 2

Ⓑ For Changes see Sheets 1, 2 and 3.

APPROVED MARCH 4 1954 REVISED Ⓐ 28 APRIL 63 Ⓑ 9 JUNE 1967

REVIEWER: MI, EL, ME, VC, SM
USER: OS, MC, GL, AT, MU

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A VC Other Cust SH	TITLE SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), MEDIUM CARBON STEEL, PLAIN FINISH, UNC-2A	MILITARY STANDARD MS 35295	
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES	SHEET 1	OF 3

DD FORM 1 MAY 59 672-1 (Continued)

FED. SUP CLASS
5305

			Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ	Ⓑ
NOMINAL SIZE			3/4	7/8	1	1-1/8	1-1/4	1-3/8	1-1/2
THREADS PER INCH			10UNC	9UNC	8UNC	7UNC	7UNC	6UNC	6UNC
BODY DIAMETER	D	Max	.7500	.8750	1.0000	1.1250	1.2500	1.3750	1.5000
		Min	.7410	.8660	0.9900	1.1140	1.2390	1.3630	1.4980
WIDTH ACROSS FLATS	A	Max	1.1250	1.3125	1.5000	1.6875	1.8750	2.0625	2.2500
		Min	1.1000	1.2850	1.4690	1.6310	1.8120	1.9940	2.1750
WIDTH ACROSS CORNERS	W	Max	1.2990	1.5160	1.7320	1.9490	2.1650	2.3820	2.5980
		Min	1.2540	1.4650	1.6750	1.8590	2.0660	2.2730	2.4800
HEAD HEIGHT	H	Max	.4830	.5630	.6270	.7180	.8130	.8780	.9740
		Min	.4550	.5310	.5910	.6580	.7490	.8100	.9020
RADIUS	R	Max	.041	.062	.093	.093	.093	.093	.093
		Min	.021	.041	.062	.062	.062	.062	.062

LENGTH L	TOLERANCE Size 3/4 and under	TOLERANCE Size 7/8 and over	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
3/4	+0 -1/32	+0 -1/16	~~179~~						
7/8			~~180~~	~~202~~					
1			~~181~~	~~203~~	~~224~~				
1-1/8	+0 -1/16	+0 -1/8	~~182~~	~~204~~	~~225~~	~~245~~			
1-1/4			~~183~~	~~205~~	~~226~~	~~246~~	~~265~~		
1-3/8			~~184~~	~~206~~	~~227~~	~~247~~	~~266~~	~~284~~	
1-1/2			~~185~~	~~207~~	~~228~~	~~248~~	~~267~~	~~285~~	~~302~~
1-3/4			~~186~~	~~208~~	~~229~~	~~249~~	~~268~~	~~286~~	~~303~~
2			~~187~~	~~209~~	~~230~~	~~250~~	~~269~~	~~287~~	~~304~~
2-1/4	+0 -3/32	+0 -3/16	~~188~~	~~210~~	~~231~~	~~251~~	~~270~~	~~288~~	~~305~~
2-1/2			~~189~~	~~211~~	~~232~~	~~252~~	~~271~~	~~289~~	~~306~~
2-3/4			~~190~~	~~212~~	~~233~~	~~253~~	~~272~~	~~290~~	~~307~~
3			~~191~~	~~213~~	~~234~~	~~254~~	~~273~~	~~291~~	~~308~~
3-1/4			~~192~~	~~214~~	~~235~~	~~255~~	~~274~~	~~292~~	~~309~~
3-1/2			~~193~~	~~215~~	~~236~~	~~256~~	~~275~~	~~293~~	~~310~~
3-3/4			~~194~~	~~216~~	~~237~~	~~257~~	~~276~~	~~294~~	~~311~~
4			~~195~~	~~217~~	~~238~~	~~258~~	~~277~~	~~295~~	~~312~~
4-1/4			~~196~~	~~218~~	~~239~~	~~259~~	~~278~~	~~296~~	~~313~~
4-1/2			~~197~~	~~219~~	~~240~~	~~260~~	~~279~~	~~297~~	~~314~~
4-3/4			~~198~~	~~220~~	~~241~~	~~261~~	~~280~~	~~298~~	~~315~~
5			~~199~~	~~221~~	~~242~~	~~262~~	~~281~~	~~299~~	~~316~~
5-1/2			~~200~~	~~222~~	~~243~~	~~263~~	~~282~~	~~300~~	~~317~~
6			~~201~~	~~223~~	~~244~~	~~264~~	~~283~~	~~301~~	~~318~~

NOTE: For illustration, material, finish and other pertinent data, see sheet 1.

Ⓑ CANCELLED IN PART AFTER 9 June 1967. USE MS90725.
INTERCHANGEABILITY RELATIONSHIP - MS90725 can replace the cancelled screws of the same size.
The cancelled MS35295 cannot always replace MS90725 and should be used until existing stock is depleted.
Then use only the superseding MS screws of the same size for design and replacement.

Ⓑ CANCELLED IN PART
SEE NOTE

REVIEWER: MI, EL, ME, WC, SH
USER: OS, MC, GL, AT, MU

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED MARCH 4, 1954 REVISED Ⓐ 26 APRIL 63 Ⓑ FOR CHANGES SEE SHEETS 1,2 AND 3

P.A. WC Other Cust SH	TITLE SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), MEDIUM CARBON STEEL, PLAIN FINISH, UNC-2A	MILITARY STANDARD MS 35295
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 2 OF

DD FORM 1 SEP 57 672-1 (Continued)

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FED. SUP CLASS
5305

REVIEWER: MI, EL, ME, WC, SH
USER: OS, MC, GL, AT, MU

			Ⓑ	Ⓑ	Ⓑ	Ⓑ			
NOMINAL SIZE			1-3/4	2	2-1/4	2-1/2			
THREADS PER INCH			5UNC	4-1/2UNC	4-1/2UNC	4UNC			
BODY DIAMETER	D	Max Min	1.7500 1.7380	2.0000 1.9880	2.2500 2.2380	2.5000 2.4880			
WIDTH ACROSS FLATS	A	Max Min	2.6250 2.5380	3.0000 2.9000	3.3750 3.2620	3.7500 3.6250			
WIDTH ACROSS CORNERS	W	Max Min	3.0310 2.8930	3.4640 3.3060	3.8970 3.7190	4.3300 4.1330			
HEAD HEIGHT	H	Max Min	1.1340 1.0540	1.2630 1.1750	1.4230 1.3270	1.5830 1.4790			
RADIUS	R	Max Min	.093 .062	.093 .062	.093 .062	.093 .062			
LENGTH	L	Tolerance	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number	Dash Number
1-3/4		+0 -1/8	~~319~~						
2			~~320~~						
2-1/4		+0 -3/16	~~321~~	~~335~~					
2-1/2			~~322~~	~~336~~					
2-3/4			~~323~~	~~337~~	~~349~~	~~361~~			
3			~~324~~	~~338~~	~~350~~	~~362~~			
3-1/4			~~325~~	~~339~~	~~351~~	~~363~~			
3-1/2			~~326~~	~~340~~	~~352~~	~~364~~			
3-3/4			~~327~~	~~341~~	~~353~~	~~365~~			
4			~~328~~	~~342~~	~~354~~	~~366~~			
4-1/4			~~329~~	~~343~~	~~355~~	~~367~~			
4-1/2			~~330~~	~~344~~	~~356~~	~~368~~			
4-3/4			~~331~~	~~345~~	~~357~~	~~369~~			
5			~~332~~	~~346~~	~~358~~	~~370~~			
5-1/2			~~333~~	~~347~~	~~359~~	~~371~~			
6			~~334~~	~~348~~	~~360~~	~~372~~			

NOTE: For illustration, material, finish and other pertinent data, see sheet 1.

Ⓑ CANCELLED IN PART
SEE SHEET 2

APPROVED MARCH 4, 1954 REVISED Ⓐ 26 APRIL 63 Ⓑ FOR CHANGES SEE SHEETS 1, 2 AND 3

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P.A. WC Other Cust SH	TITLE SCREW, CAP, HEXAGON HEAD (FINISHED HEXAGON BOLT), MEDIUM CARBON STEEL, PLAIN FINISH, UNC-2A	MILITARY STANDARD MS 35295
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 3 OF

DD FORM 1 SEP 57 672-1 (Continued)

*U.S. GOVERNMENT PRINTING OFFICE: 1967-201-500/107

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